A Novel Framework for Urban Land Cover Change Detection With NASA's Black Marble Nighttime Lights Product

Yuxin Zhang, Xuecao Li, Jianxi Huang, *Senior Member, IEEE*, Haixiang Guan, and Hai Huang

***Abstract*— Against rapid development in urban areas, timely urban land cover change (ULCC) information is beneficial for understanding the urban environment and promoting sustainable development. To realize real-time ULCC detection, high-frequency remotely sensed data are urgently needed. In this study, we tested the detection capability of ULCCs using a new daily nighttime light (NTL) image (Black Marble). First, time series of VNP46V2 from 2012 to 2019 was collected and decoded into annual trend segments using the BFAST Monitor model. Then, we recognized the jump point in trend segments and defined the corresponding pixel as ULCC. We analyzed the normalized difference vegetation index (NDVI) time series from Landsat images spanning 2014–2019 and removed pixels without significant seasonal fluctuations from ULCC assembled. Finally, the magnitude and change time of ULCC pixels were quantified through the BFAST Monitor. It was proved that Black Marble performed well in ULCC detection, achieving an overall accuracy of 87.75%, and detected change time was accurate to 81.12% under ±1 year allowable error. The present Black Marble data have the potential for real-time urban land use detection and global mapping of ULCC, especially in areas without enough clear-sky observations.**

***Index Terms*— Black Marble, land change detection, nighttime light (NTL), urban land cover change (ULCC).**

I. Introduction

URBAN land cover changes (ULCCs) are human activities' responses to global environmental changes and socioeconomic transformation [1], [2], [3]. Global rapid urbanization will lead to massive fragmentation of natural habitats by 2100 [4] and a 1%–4% decline in crop production [5]. Therefore, keeping up with the latest dynamic ULCC information is crucial for understanding urban heat islands, temperature change, and environmental issues [6], [7], [8], [9], [10], and supporting urban planning and monitoring the land construction process [11].

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ULCC engraves urban areas undergoing eternal artificial modification, from water bodies and vegetation to impervious surfaces [12], [13], urban renewal, including reconstruction for new buildings, and demolition into bare land or green space [14], [15]. The process of urban land use changes complexly along with human activity dynamics, which is hard to investigate in the field. Remotely sensed images have demonstrated overall success in ULCC detection [16], [17] and published many widely appreciated datasets, such as GAIA [18], NUACI [19], and GHSL [20], from coarse spatiotemporal mapping to finer ones [21]. The detection of ULCC is a subject of great attention [22], [23], which can be divided into two categories according to the study framework. The first is postclassification change detection [24], [25]. According to the spectral feature or its ramifications, such as the normalized difference vegetation index (NDVI) [26], [27], it is simplified to identify impervious surfaces from the permeable mainly from physical attributes and not be assumed to reverse. We can recognize changes between labeled pixels of postclassification maps of different time stamps [28].However, it impedes free-frequency change detection, especially in underdeveloped countries [29], and it is hard to quantify change magnitude and refine change time. The other category can be summarized as time-series change detection [30], [31], which searches for abrupt and gradual breaks in remotely sensed time series and emphasizes time, magnitude, and duration of change [32]. Several valuable change detection algorithms have been developed and serve as great opportunities for ULCC, such as Landtrendr [33], CCDC [34], CUSUM [35], and BFAST [36], and BFAST Monitor has great performance on accuracy and running speed [37]. Remotely sensed time series from the Landsat is one of the most important sources for understanding ULCC [38], [39]. For this kind of data overlapping the exact location every 16 days, detecting abrupt and gradual breaks is applicable, such as fire and urban expansion [40], [41]. However, sufficient observations are demanding when utilizing a change detection algorithm, but the optical sensor is sensitive to climate conditions, and generally, massive pixels impacted by the cloud cannot be used for analysis. Higher time resolution (e.g., daily) is needed to provide more time stamps than Landsat in the same time gap. Moreover, nighttime light (NTL) data provide a novel perspective of land use change measurement, which could measure human activity signals through nightlights that Landsat does not capture.

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National Aeronautics and Space Administration (NASA) moonlight and atmosphere Black Marble product (VNP46A2) collected by Visible Infrared Imaging Radiometer Suite/NPOESS Preparatory Project (VIIRS/NPP) [42], a kind of daily NTL data that may be beneficial for timely monitoring the complex and changeable urban environment. Black Marble products compensate for aerosols, water vapor, and ozone impacts on the NTLs, remove biases introduced by extraneous sources of NTL emissions, and correct uncertainties stemming from seasonal variations. NTL long-term time series efficiently measures artificial light at night from space [43], [44], visually reflecting human activities' socioeconomic attributes and providing unique information to understand urban changes. NTL has perceived the changing world from human activities and urban environmental impacts [45], as the primary information for monitoring the urbanization process, especially mapping impervious surface area from urban areas [46], [47], [48], estimating population and gross domestic product (GDP) [49], [50], evaluating social issues (i.e., poverty) [51], detecting public incidents (i.e., war and epidemic) [52], [53], and assessing electric power consumption and carbon dioxide (CO_2) emissions [54], [55]. NTL data have been frequently used to map urban land use through threshold-based, machine-learning, and deep-learning methods. For instance, Wang and Shen [56] applied an analyzing framework combing kernel density estimation (KDE) under different search radii and threshold-based extraction to detect the boundary of urban areas. Liu et al. [57] mapped urban extent in the conterminous United States (CONUS) using machine learning methods with VIIRS/Night Band (DNB). He et al. [58] proposed a fully convolutional network (FCN) integrating multiple remotely sensed sources to extract the global urban land from 1992 to 2016. However, NTL data in urban land use detection still needs to be improved. Widely used data such as the Defense Meteorological Satellite Program's Operational Line Scanner (DMSP/OLS) perform noticeable blooming effects that lead to confusing urban boundary extraction. Besides, NTL's monthly or annual synthetic data are used dominantly to reduce data variation, making it hard to detect short-term urban land use changes promptly. The high-frequency NTL images Black Marble product (VNP46A2) for urban land use changes analysis are urgently needed to track real-time ULCC.

Here, we tested a new high-frequency NTL image for ULCCs analysis using BFAST Monitor, a change detection method decoding NTL series and gaining annual trend segments. Under the NDVI time-series supplement, we mathematically extracted ULCC and quantified their magnitude and change time. Then, we aim to prove Black Marble's ability in ULCC detection, which provides a new approach for real-time urban detection and comparative studies.

II. Study Area and Data

A. Study Area

This study selected five patches in Beijing, China (39°26′N–41°03′, 115°25′E–117°30′E) as the study area [Fig. 1(a)]. Beijing city is a collection of vast types and phenomena of urban change, as it has undergone the most dramatic land use and land cover changes in the past few decades. Five patches in an area of 10 km^2 were randomly selected as main study areas to detect ULCC, and each patch was divided into 20 × 20 units (500 m) to match the resolution of the NTL images [Fig. 1(b)]. These patches include both developed urban core areas and emerging urban areas on the edge of the city. Therefore, we test a novel type of high-frequency NTL images for ULCC detection and analysis.

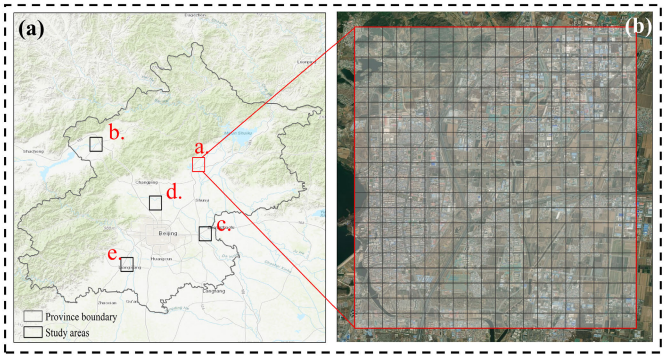


Fig. 1. (a) Five patches for ULCC with an area of 10 km^2, and (b) each patch is divided into 20 × 20 units to match the resolution of the NTL images.

B. Data Collection and Processing

This study mainly used daily NTL from Black Marble to detect ULCC. NASA's Black Marble NTLs product suite (VNP46) has been available at 500-m resolution since January 2012 with data from VIIRS/DNB onboard the Suomi National Polar-orbiting Platform (SNPP) [42]. It provides high-quality nightly NTL characterization based on cloud-free, atmospheric terrain, vegetation, snow, lunar, and stray light-corrected DNB radiance. Asymmetric Gaussian fitting is employed to fill the temporal gaps of the VNP46A2, This product enables abrupt, seasonal, and gradual detection of all NTL-related changes at the native DNB pixel scale. Therefore, we used VNP46A2, a moonlight-adjusted NTL product, to detect long-term ULCCs. Gap-filled DNB BRDF-Corrected NTL layers were used, and pixels with 65 535 value in time series were omitted from the analysis. Adjusted NTLs yearly product (VNP46A4) are collected to discuss the influence of the blooming effect on this study, which is caused by the mutual influence of lights from a pixel and its neighbors.

We utilized Landsat surface reflectance images as our secondary datasets covering the year 2014–2019 from the sensor Operational Land Imager (OLI). All images were acquired from Google Earth Engine (GEE) and were used to calculate the NDVI, based on an equation between the red band (0.64–0.67 um) and near infrared band (0.85–0.88 μm), on account of the NDVI of vegetation was higher than that of impervious surface. NDVI time series was sorted by day of the year (DOY) as a supplementary dataset to exclude significant urban change but non-ULCC pixels. To match the resolution of VNP46A2, we took the average of all pixels in each 500-m unit.

III. Methodology

This study aims to detect ULCC using novel night light data (Fig. 2). We assume that ULCC must accompany long-term

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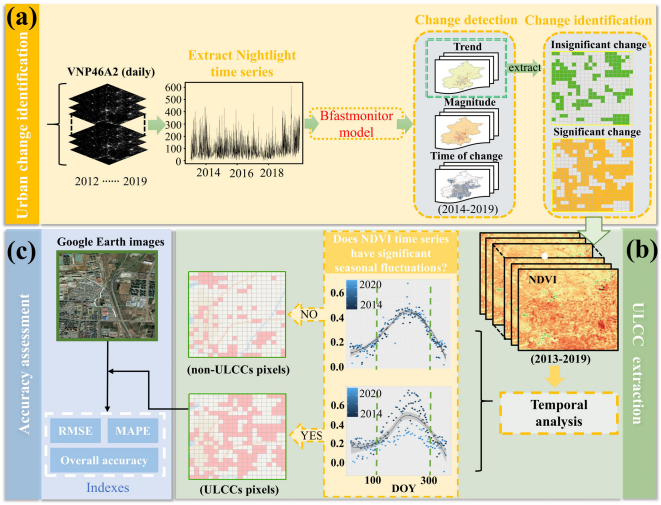


Fig. 2. Framework for urban land change analysis: (a) adapting daily NTL data (VNP46A2) to identify significant urban change, (b) utilizing Landsat data (NDVI) to extract ULCCs, and (c) accuracy assessment of ULCC and the corresponding time of change.

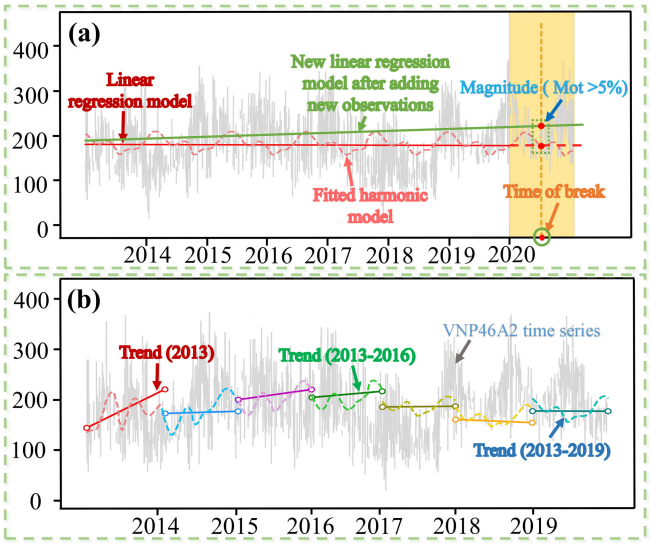


Fig. 3. BFAST Monitor used for decoding VNP46A2 data and extracting urban changes. (a) Season-trend model including linear regression model based on stable history data (the red line), fit harmonic model (the dashed red curves), and new linear regression model (the green line) after adding new observations where there are shades of orange. The magnitude of the detected break describes the median of the difference between the data and the model prediction in the monitoring period (the middle and the upper red dot) when MO_t is greater than 5% probability and its corresponding time of break (the lower red dot) can be obtained. (b) Trend segments to identify urban changes (solid lines).

noticeable NTL changes. Therefore, the core idea stands on detecting significant breaks among historical data. First, we utilized a change detection algorithm (BFAST Monitor) to decode daily NTL time series (VNP42A2) from 2012 to 2019 and obtain yearly trend segments, change magnitude, and time of change [Fig. 2(a)]. Then, based on annual trend segments, we can judge if there exists a significant break in each year. However, significant changes may be mixed with pure human activity changes without land cover changes. Therefore, we adapted the NDVI time series as supplemented data to exclude pure human activity change pixels [Fig. 2(b)]. ULCC are seasonal fluctuations in the NDVI time series. Finally, this study estimated its accuracy by several indexes [Fig. 2(c)]. Ground truth labels and the corresponding time of change were collected from Google Earth Images.

A. Decoding the High-Frequency Nighttime Light Data

BFAST Monitor model [36], a multipurpose time-series-based disturbance detection approach, was utilized for decoding VNP46A2 data and extracting trend segments, magnitude, and time of break each year. BFAST Monitor fits stable historical observations to enable change detection within newly acquired data (Fig. 3). Based on stable history observation y_t at time t ($t = 1, \ldots, n$), a season-trend model is assumed with a linear regression model and fit harmonic model [Fig. 3(a)]. The season-trend model is described as follows:

$$y_t = \alpha + \beta t + \sum_{j=1}^{k} \gamma_j \sin\left(\frac{2\pi j t}{f} + \delta_j\right) + \varepsilon_t \quad (1)$$

where intercept α, trend β and amplitude γ_j and season phase δ_j are estimated using ordinary least squares (OLS). Harmonic term k is 3, and higher value represents variations that occur on a three-month cycle or less [59], but we focus on annual changes. f is the frequency of observations which is set to 365. ε_t is the residual term at time t.

Based on the fit season-trend model, moving sums of the residuals (MO_t) measuring if it remains stable for the newly required observed period ($t = n+1, \ldots, m$) is used to define the break (2), aiming to check whether season-trend model estimated from history period still fit new observations. If so, the new data are consistent with the history

$$MO_t = \frac{1}{\hat{\sigma}_0\sqrt{n}} \sum_{s=t-h+1}^{t} (y_s - \hat{y}_s + \varepsilon_s) \quad (2)$$

where $\hat{\sigma}_0$ is the standard deviation of ε_t from the history period, and $h = n/4$. $\hat{y}_s$ is the season-trend model with consistently estimated $\hat{\alpha}_1$, $\hat{\alpha}_2$, $\hat{\gamma}_k$, and $\hat{\delta}_k$. If new observations remain stable, the MO_t should be close to zero or fluctuate randomly in a small range. Otherwise, it will change regularly from zero until its absolute value is greater than 5% probability under structural stability when clarified to be a break-point.

The magnitude quantified the degree of detected change that may demonstrate the intensity of human activity change. It is derived from the difference between trend components when the break happens

$$\text{Magnitude} = (\alpha_j - \alpha_{j-1}) + (\beta_j - \beta_{j-1})t_0 \quad (3)$$

where α_{j-1}, α_j, β_j, and β_{j-1} mean intercept α and the trend β before and after the break point j at the time of break t_0.

In this study, the BFAST Monitor model is constructed sequentially with one year as the step size to extend the history data size. The start of historical observations is in 2013, and the end is the year 2019. The trend can be extracted from the BFAST Monitor model, which can be estimated and tested using OLS. Fig. 3(b) shows trend segments from trend_{2013} ($t = 2013$) to trend_{2019} ($t = 2013, \ldots, 2019$). It is equal to

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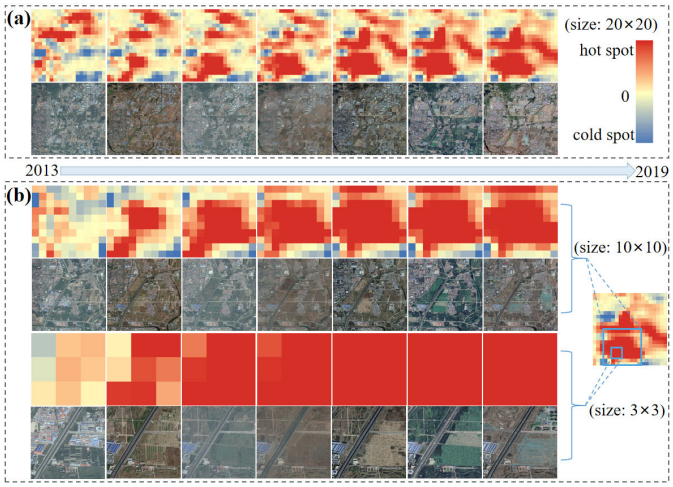


Fig. 4. Trend segments of a patch (a) in different sizes from 2013 to 2019, including (a) 20 × 20 units and (b) 0 × 10, and 3 × 3 units. The warm color areas mean hot spots with stronger changes in human activity, and the cold color areas refer to cold spots where human activity is stable.

the slope α in the trend-season model. We increased the length of stable history inputs year by year, so we can believe that the new year has a human activity fluctuation if its trend has a jump compared with the previous one. Temporal and spatial variation of trend segments can be the hot and cold spots areas of human activity in a city, which can be considered a tentative and hanging sign of urban change (Fig. 4).

B. Extraction of Urban Land Cover Changes

In this study, trend segments are regarded as important material. The time of change detected by the BFAST Monitor model could have been directly used as the change year, but it has the disadvantage of overestimation and will include pseudo multiple breakpoints in this study, so we combined standard error as a priority. The standard error of trend segments σ is constructed to quantify the variation of trend across n years

$$\sigma = \sqrt{\frac{1}{n \times (n-1)} \sum_{1}^{n} \text{trend}_i^2} \tag{4}$$

where trend_i means the difference between trend at year i and the average across n years. It reflects the variation in the trend gatherings.

Δtr means the difference of trend between adjacent two time period (5). If the trend segments remain stable, equal to steady NTL images time series, we tested σ as a threshold (Fig. 5) and Δtr should be close to σ and fluctuates within σ. we assume that if Δtr is greater than σ, there is a ULCC [Fig. 6(a)]

$$\Delta\text{tr} = \text{trend}_{i+1} - \text{trend}_i. \tag{5}$$

It is assumed that ULCC will undoubtedly cause a significant NTL change, and non-ULCC has it on both sides. Therefore, urban land can be classified into three classes [Fig. 6(b) and (c)].

1) *Class 1:* Significant urban change and ULCC ($\Delta\text{tr} > \sigma$), including urbanization, urban renewal, and counter-urbanization.

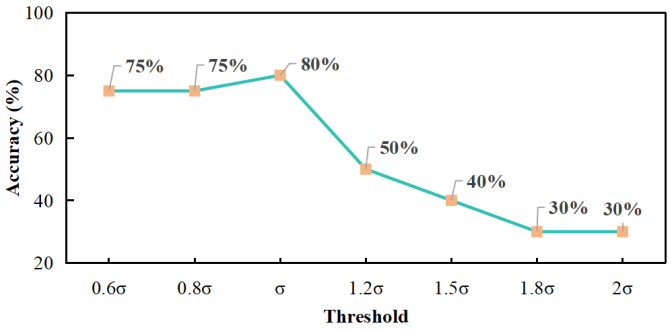


Fig. 5. Sensitivity of the threshold σ, which has the highest overall accuracy could balancing the commission and omission of detection.

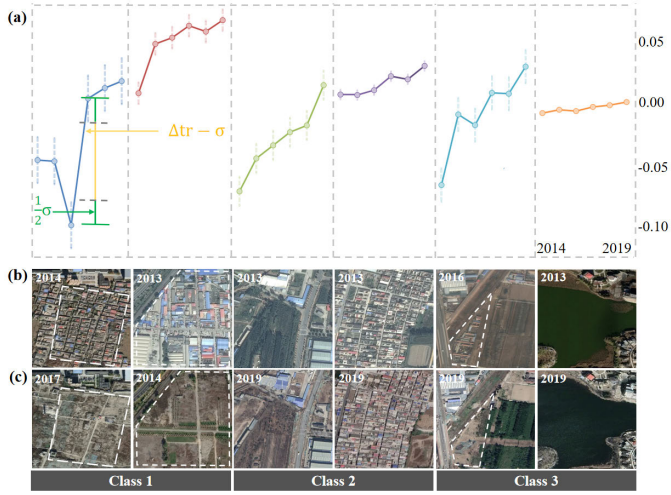


Fig. 6. Typical ULCCs and their characteristics, including three classes named: 1) significant urban change and ULCC; 2) significant urban change but non-ULCC; and 3) insignificant urban change and non-ULCC. (a) Standard error (σ) of trend segments (dashed lines) and the judging criterion of significant urban change (Δtr-σ) (orange lines). (b) Surface before changes. (c) Surface after changes.

2) *Class 2:* Significant urban change but non-ULCC ($\Delta\text{tr} > \sigma$). The trend jump comes from the increasing intensity of human activity.
3) *Class 3:* Insignificant urban change and non-ULCC ($\Delta\text{tr} < \sigma$). Urban land cover remains the same or changes slightly, and the intensity of human activity is immobile.

We can exclude part of non-ULCC (class 3) based on the relationship between Δtr and σ [Fig. 7(a)]. Class 1 and class 2 perform similarly in NTL but not in NDVI. Suppose the NDVI time series ranged by DOY shows significant seasonal fluctuation. In that case, it will be summarized as ULCC [Fig. 2(b)]. In this way, three classes and the ULCC can be distinguished [Fig. 7(b)]. We assume that ULCC will cause the highest magnitude of NTL, that the maximal magnitude corresponding time of change is considered as happening construction period.

C. Accuracy Assessment

In this study, we adapted five metrics to evaluate the performance of ULCC detection [60]. The accuracy (ACC) shows all correctly classified samples proportion (6). The

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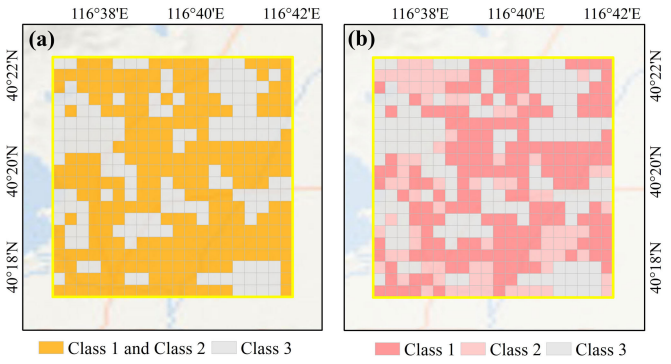


Fig. 7. Taking patch (a) as an example: (ais the result of trend segments variation analysis that distinguishes significant urban change (Class 1 and Class 2) and insignificant urban change (Class 3) and (b) extraction of ULCC (Class 1) and non-ULCC (Class 2).

positive predictive value (PPV) means the correctly classified proportion of predicted ULCC (7). The true positive rate (TPR) stands for the correctly classified proportion of labeled ULCC (8), and true negative rate (TNR) is the correct classified proportion of labeled non-ULCC (9). $F1$ Score is a weighted average of PPV and TPR (10)

$$\mathrm{ACC} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}+\mathrm{TN}} \tag{6}$$

$$\mathrm{PPV} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{7}$$

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{8}$$

$$\mathrm{TNR} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}} \tag{9}$$

$$F1 = \frac{2\mathrm{TP}}{2\mathrm{TP}+\mathrm{FN}+\mathrm{FN}} \tag{10}$$

where TP, TN, FP, and FN are the number of true ULCC, ture non-ULCC, false ULCC and false non-ULCC samples.

Meanwhile, root-mean-square error (RMSE) (11), mean absolute error (MAPE) (12), and overall accuracy (the proportion of correct extraction) (13) are used to evaluate the result of the detected time of change. All of the ground truth comes from Google Earth's high-resolution images. Regarding the coarse time resolution of Google Earth images and the construction period, the allowable error is set as ± 1 year

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{11}$$

$$\mathrm{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{\hat{y}_i - y_i}{y_i}\right| \tag{12}$$

where $\hat{y}_\mathrm{i}$ and y_i denote the detected label and the ground truth, respectively,

$$\text{overall accuracy} = \frac{N}{M} \times 100\% \tag{13}$$

where N and M stand for the correct detection samples number and the total number of samples for accuracy assessment.

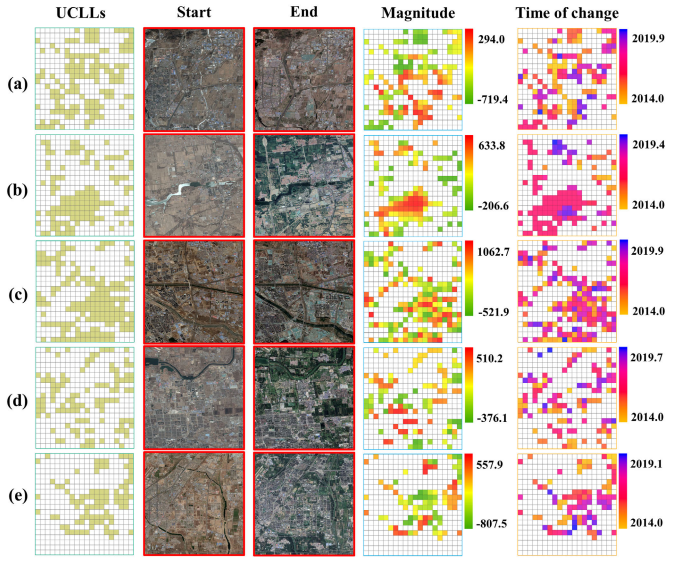


Fig. 8. Mapping performance of ULCC detection at study areas [patch (a)–(e)], including the spatial patterns, magnitude, and detected change time of UCLL, and the true surface reference from Google Earth images mainly about 2013 as start year and 2019 as end year.

IV. Results and Discussion

A. Performance of ULCC Detection

During the detection period from the start year 2013 to the end year 2019, ULCC in patches (a–e) were identified (Fig. 8). From the spatial patterns of ULCC, most of ULCC merge into a block over time, and the remaining scattered plot features urban renewal especially demolition. Demolition is always accompanied by NTL disappearance with a negative magnitude. However, rebuilding is meant to seek sufficient land use, gathering more NTL and positive magnitude to ensure economic and intensive land use. Urban expansion generates NTL and has a positive magnitude either. Therefore, the magnitude can be used as a basis for judging ULCC types. Urban expansion and rebuilding have an aggregated magnitude pattern, but demolition is separated. According to the magnitude and time of change, the evolution pattern of urban expansion and rebuilding also can be summarized. For example, given the time of ULCC and adopting the definition from Liu et al. [61], patch (b) can be classified as infilling from the edge to the center, and patch (c) is a typical outlying ULCC. Therefore, the magnitude and time of change are greatly useful for understanding urban change.

B. Accuracy Assessment of ULCC Detection

Assessing the ULCC detection accuracy with true labels from Google Earth images, the confusion matrix of study patch (a) is shown in Fig. 9(a). Overall, the identification of ULCC and non-ULCC performs well, with 82% and 84% correctly classified pixels, but 18% and 16% pixels misidentified. Comprehensively, five patches have an effective ACC between 78.75% and 87.75%, with a median $F1$ above 82% [Fig. 9(b)]. According to PPV, 70% to 85% of predicted ULCC are correct. About 82.44% and 84.03% of true ULCC and non-ULCC were precisely detected following TPR and TNR. Five patches

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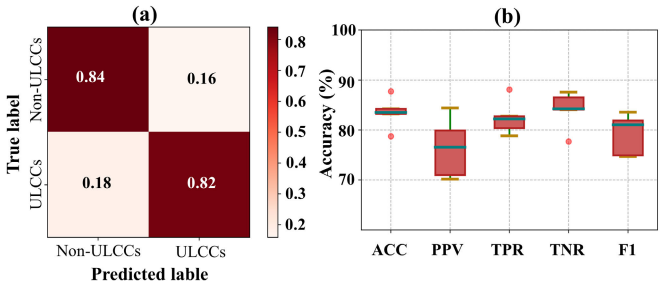


Fig. 9. Accuracy assessment of ULCC detection. (a) Confusion matrix of the patch (a), obtaining actual labels from Google Earth images. (b) Comprehensive accuracy assessment covering study areas [patch (a)–(e)] containing ACC (accuracy), PPV (precision), TPR (sensitivity), TNR (specificity), and F1.

distributed in different places were randomly selected. Their highly concentrated distribution of accuracy indexes illustrates the objectivity and reliability of ULCC detection results.

C. Accuracy Assessment of Detected Change Time

Using ground truth labels from Google Earth images as the reference, we assessed the accuracy of the detected change time (Fig. 10). We believe that ULCC will cause drastic NTL fluctuation so that the detected change time follows the maximum magnitude. Fig. 10(a)–(e) shows precise patch (a–e) results, with the accuracy of change time ranging from 72.07% to 93.7%. The accuracy means the proportion of the positive detected year, and the allowable error is ±1 year regarding the coarse time resolution of Google Earth images and the construction period. RMSE and MAPE estimate ULCC time accurately to month. RMSE is mainly greater than one year and up to two years, and MAPE is about 5%. Patch (b) and patch (e) perform better than the other three patches, with most pixels spreading over the 1:1 line but other patches deviating. The principal reason for this error is that patch (b) and patch (e) have an outbreak period of ULCC, which happened in most pixels simultaneously. For example, most patch (b) pixels underwent ULCC in 2019, and patch (e) happened in 2014 and 2015. It proves that regional ULCC enable this kind of high-frequency NTL data with detection algorithm performance to be fully exploited. The accuracy and MAPE of the five patches are 81.12% and 4.28%, respectively. However, their RMSE is relatively poor, up to 1.53 years [Fig. 10(f)]. According to the results of five patches, patch (b) which is farthest from the urban center experiences the highest accuracy and patch (d) located in the urban core is the worst. Generally speaking, the method in the urban center performs better than the outskirts, since land use changes in inner cities are much more complex than that on the edges.

D. Errors From Different Types of ULCC

The effect of ULCC identification extensively changes between different land cover change types. Considering the transaction of urban and nonurban land cover, primary ULCC types contain urbanization, counter urbanization, and urban renewal, which contribute to omission errors classified into the urban constant surface. As shown in Fig. 11, most omission errors derive from urbanization and urban renewal areas,

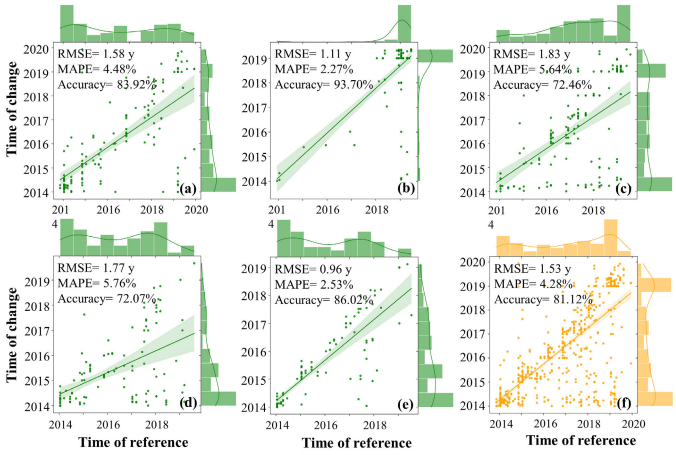


Fig. 10. Accuracy assessment of detected change time. (a)–(e) Linear fitting between the detected time of change and time of reference from Google Earth. (f) Synthesis of (a)–(e). The surrounding frequency maps of each subfigure indicate the detected and reference change time.

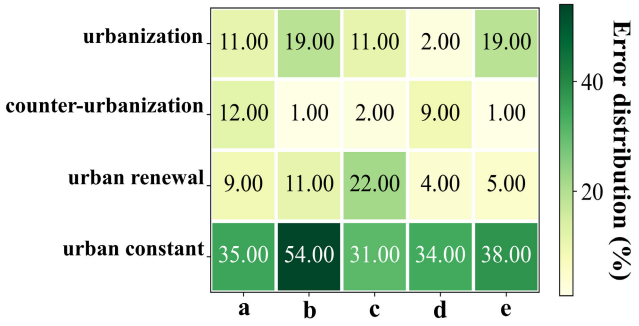


Fig. 11. Errors of ULCC extraction distributed in different land cover change types in the patch (a)–(e), including omission errors (urbanization, counter urbanization, and urban renewal) and commission errors (urban constant areas).

accounting for 10%–20%. There are some typical sources: a high coverage of urban greening that weakens NDVI change magnitude caused by demolition and construction, especially in areas with sparse vegetation; high housing vacancy that lacks a significant break in NTL time series; short construction duration that is missed in the seasonal testing circle; informal settlements with no NTL single. Counter-urbanization areas rarely were omitted, for the signal is clear that NTL suddenly goes from presence to absence, only coming in at 5% medially. However, urban constant areas extracted as ULCC are considered the most influential error, up to 30%–55%. Two aspects of physical fluctuation cause it. The one is NDVI, such as crop rotation and water conservancy facilities transformed from farmland. The other is that NTLs, such as farmland or forest, are modified and endowed with urban green spaces.

Because of different construction features, the ability to change time detection differs from ULCC types (Fig. 12). Through KDE mapping, the consistency between the time of detection and time of reference is clarified. Overall, the performance of urbanization areas is greater than urban renewal and counter-urbanization, with an accuracy of 90.22%. RMSE and MAPE were relatively low, 1.16 years and 2.85% [Fig. 12(a)]. However, urban renewal [Fig. 12(b)] and counter urbanization

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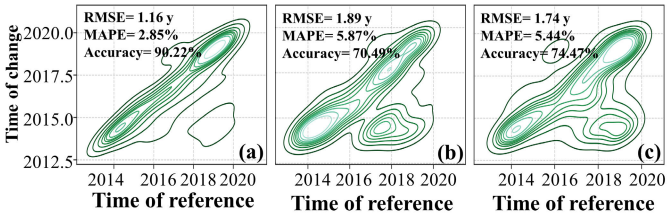


Fig. 12. Accuracy assessment of detected change time in different ULCC types, including: (a) urbanization, (b) urban renewal, and (c) counter urbanization. Mapping is based on kernel density estimation, so the green curve is the kernel density curve. The smaller are the intervals between kernel density curves, the higher is the consistency between the time of detected change and reference.

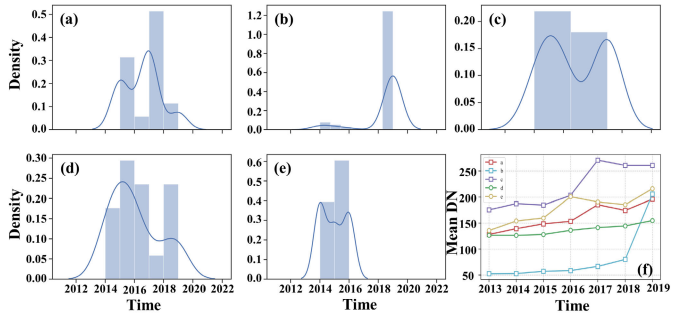


Fig. 13. Detected significant change year in commission error pixels. (a)–(e) Density of detected time in commission error pixels (urban constant) from patch (a) to (e), respectively. (f) Annual mean DN of the patch (a)–(f) from 2013 to 2019.

(Fig. 12) were far less successful than expected. Their accuracy decreased to 70.49% and 74.47%, respectively, and RMSE was close to two years. Even if MAPE was considerable, reaching 5.87% and 5.44%.

E. Errors From the Blooming Effect of Nighttime Light Data

Except for the errors caused by the surface features, the blooming effect of NTL data is also responsible. Theoretically, the trend segments of non-ULCC do not have a significant jump, so they will be classified into urban constant areas. However, parts of pixels will be radiated by surroundings drastically at NTL changes. At the same time, the characteristic of NDVI is fuzzy and defined as ULCC eventually. To test this prediction, we counted the annual mean DN of patches from 2013 to 2019 and comparing with a detected year of ULCC in commission errors pixels (Fig. 13). The inflection year of annual mean DN [Fig. 13(f)] shows remarkable consistency with a patch (a), patch (b), and patch (e) [Fig. 13(a), (b), and (e)], respectively corresponding to 2017, 2019, and 2016. The annual mean DN of the patch (d) remains stable, and the detected year of ULCC evenly distributed [Fig. 13(d)], showing good agreement. Therefore, it proves that ULCC also influenced by the blooming effect.

V. Conclusion

In this study, we tested the ability of novel high-frequency NTL images for ULCC analysis. First, time series of VNP46V2 is collected and decoded into annual trend segments using a harmonic model named BFAST Monitor. Then, the break year can be distinguished based on annual trend segments. Besides, NDVI time series from Landsat were grouped to remove poor human activity changes without land cover changes. Finally, ULCC pixels were extracted according to break year; their magnitude and change time were also quantified. The performance of VNP46V2 identifying ULCC achieves an overall accuracy of 87.75%, and change time detection is up to 81.12%.

This study uses new daily night light data to test whether it can detect urban land. Urban development is witnessing rapid urban land cover and structure changes, especially under the drive-by land policy, which can be visually captured at night. Many studies have discussed urban issues using NTL data such as DMSP/OLS images [62], [63]. However, limited by the satellite revisiting period, spatial-oriented methods, such as thresholding [43], are widely used to recognize urban areas [48]. Black Marble data enable us to study cities from a temporal-oriented perspective and have the following potentials: 1) it is available in near-real time detection for it has rich data size to fit history model and real-time observations accurate to the day; 2) it has a finer temporal revolution that is beneficial for areas without enough clear-sky 16-day Landsat observations; and 3) it can be used for multiple frequency urban change, such as gradual migration of population and abrupt incidents as earthquake or epidemic.

In addition, this study used the BFAST Monitor algorithm, which is short in overestimating changes and not designed for multiple breakpoints [64]. It might introduce uncertainties in change detection, although this effect is minor in change year detection. Given that it is tested in five small patches, that blooming effect is exaggeratedly captured. However, Black Marble data still has the potential to map ULCC at a global scale after extremum value control and its influenced pixels window adjustment. Besides, we set the frequency of the BFAST Monitor to 365 days, which is efficient in yearly change detection. It has the potential to recognize urban land type by refining seasonal frequency, such as extracting commercial land through weekly segments because of human activities difference between weekends and working days.

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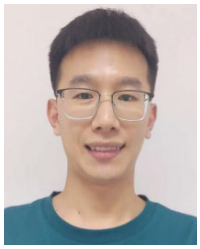
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